A01

FOUNDATION

STRUCTURE → NAME

Find the longest chain; number from the end nearer the branch

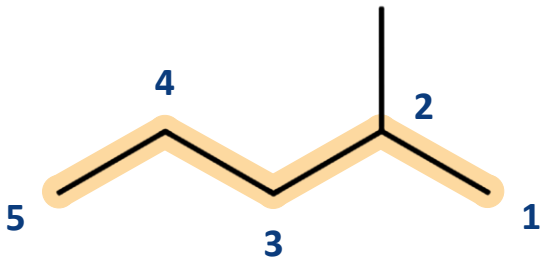
Branched C6 alkane in light naphtha and gasoline blends

MOLECULAR FORMULA

C_6H_{14}

6 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 5 C → pentane. Two equal 5-C paths exist (ending at the right-hand carbon or at the upward carbon); they give the same name.

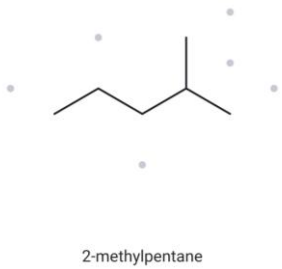
Numbering: left → right puts the methyl on C4; right → left puts it on C2. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2 → 2-methyl.

3 MY NAME

2-methylpentane

4 THE ALKANE CHECK



Returned name:

2-methylpentane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A02

FOUNDATION

STRUCTURE → NAME

The longest chain is not drawn in a straight line

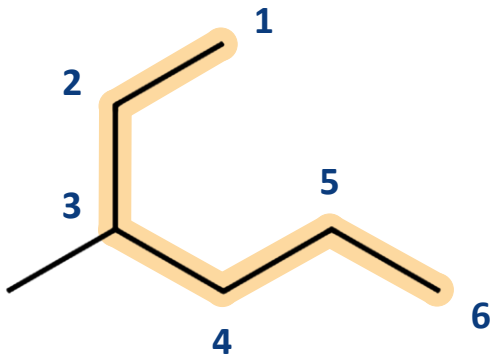
C7 gasoline-range isomer; isomer structure shifts octane number

MOLECULAR FORMULA

C_7H_{16}

7 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 6 C → hexane. It bends up through the ethyl group; the horizontal zigzag is only 5 C.

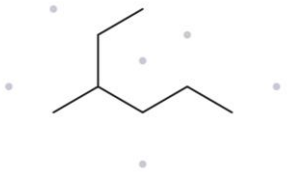
Numbering: from the top end the methyl sits on C3; from the right-hand end it sits on C4. First point of difference: 3 < 4, so number from the top end.

Substituents: methyl on C3 → 3-methyl.

3 MY NAME

3-methylhexane

4 THE ALKANE CHECK



3-methylhexane

Returned name:

3-methylhexane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A03

FOUNDATION

STRUCTURE → NAME

Two identical substituents: di- prefix, commas and hyphens

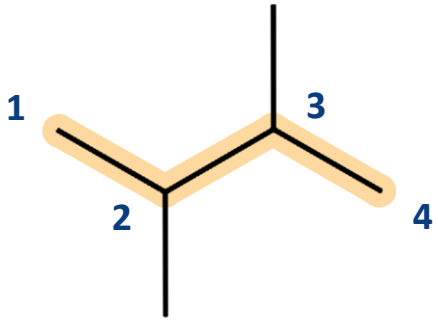
Highly branched C6 valued in isomerate (high-octane) streams

MOLECULAR FORMULA

C_6H_{14}

6 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 4 C → butane. Four equal 4-C paths exist (each uses one methyl from each branched carbon); all give the same name.

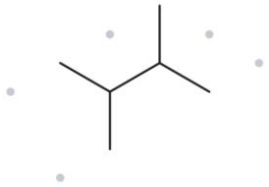
Numbering: left → right puts methyls on C2 and C3; right → left also gives C2 and C3. The locant sets 2,3 are identical, so there is no choice to make.

Substituents: two identical methyls, on C2 and C3 → 2,3-dimethyl (di- for two).

3 MY NAME

2,3-dimethylbutane

4 THE ALKANE CHECK



2,3-dimethylbutane

Returned name:

2,3-dimethylbutane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A04

DEVELOPING

STRUCTURE → NAME

Alphabetical order, not locant order, decides which substituent is cited first

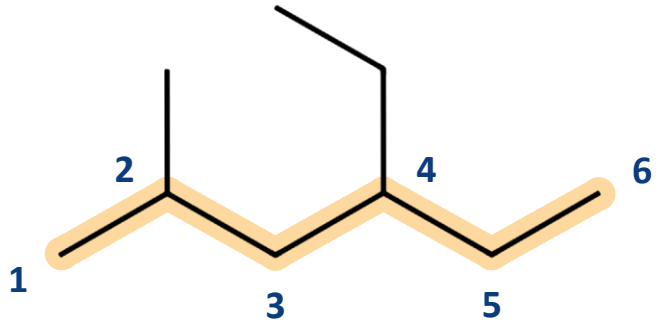
Branched C9 paraffin in the gasoline/kerosene boiling range

MOLECULAR FORMULA

C_9H_{20}

9 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 6 C → hexane. Four equal 6-C paths exist; each leaves one methyl and one ethyl as substituents, so all give the same name.

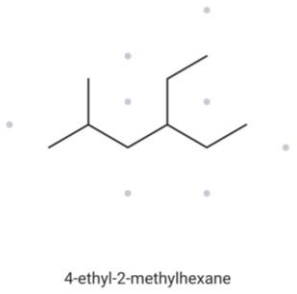
Numbering: left → right: methyl on C2, ethyl on C4 = 2,4; right → left: ethyl on C3, methyl on C5 = 3,5. First point of difference: 2 < 3, so number from the left.

Substituents: cited alphabetically: ethyl (C4) before methyl (C2), because e comes before m, even though 2 < 4 → 4-ethyl-2-methyl.

3 MY NAME

4-ethyl-2-methylhexane

4 THE ALKANE CHECK



Returned name:

4-ethyl-2-methylhexane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A05

DEVELOPING

STRUCTURE → NAME

Repeated locants and the first point of difference

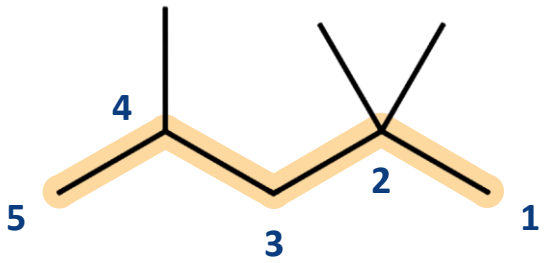
Reference compound for the gasoline octane-rating scale

MOLECULAR FORMULA

C_8H_{18}

8 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 5 C → pentane. Six equal 5-C paths exist (two possible left ends × three equivalent methyls on the quaternary carbon); all give the same name.

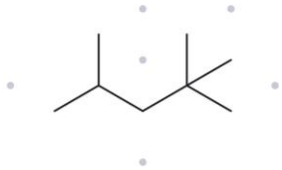
Numbering: left → right: methyls on C2, C4, C4 = 2,4,4; right → left: methyls on C2, C2, C4 = 2,2,4. First point of difference (second term): 2 < 4, so number from the right.

Substituents: three identical methyls, on C2, C2 and C4 → 2,2,4-trimethyl (tri- for three; the repeated locant is written twice).

3 MY NAME

2,2,4-trimethylpentane

4 THE ALKANE CHECK



2,2,4-trimethylpentane

Returned name:

2,2,4-trimethylpentane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A06

DEVELOPING

STRUCTURE → NAME

Halogen as a prefix; tie in locants broken alphabetically

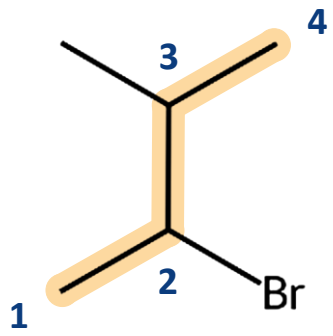
Alkyl halides are standard intermediates for substitution and elimination steps

MOLECULAR FORMULA

$C_5H_{11}Br$

5 C · open chain C_5H_{12}; Br replaces 1 H

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 4 C → butane. Two equal 4-C paths exist (the two methyls on the upper carbon are interchangeable).

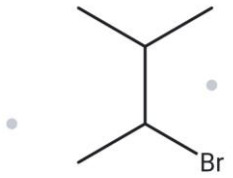
Numbering: from the lower-left CH3: bromo on C2, methyl on C3 = 2,3; from the upper-right CH3: methyl on C2, bromo on C3 = 2,3. The sets tie, so alphabetical order decides: bromo (b) is cited before methyl (m) and takes the lower locant → number from the lower-left end.

Substituents: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

3 MY NAME

2-bromo-3-methylbutane

4 THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name:

2-bromo-3-methylbutane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A07

DEVELOPING

STRUCTURE → NAME

Ring as parent; when is a locant unnecessary?

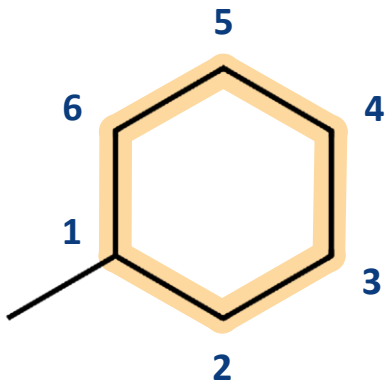
Solvent and hydrogen-carrier liquid (reversibly dehydrogenated to toluene)

MOLECULAR FORMULA

C_7H_{14}

7 C · one ring → C_nH_{2n}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: a ring is present, so the ring (6 C) is the parent → cyclohexane. The methyl is a one-carbon substituent.

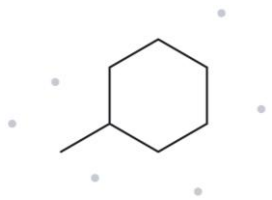
Numbering: start at the ring carbon that carries the methyl (C1). Either direction round the ring gives the same single locant 1, so there is nothing to compare.

Substituents: methyl → methylcyclohexane. A lone substituent on a ring needs no locant (not 1-methyl).

3 MY NAME

methylcyclohexane

4 THE ALKANE CHECK



methylcyclohexane

Returned name:

methylcyclohexane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A08

CHALLENGE

STRUCTURE → NAME

Ring numbering with two substituents; alphabetical tie-break on a ring

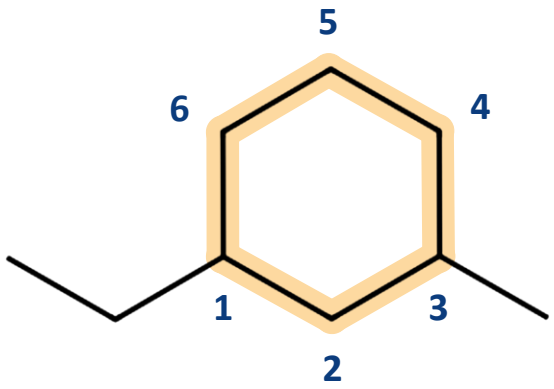
Alkylcyclohexanes (naphthenes) are a major hydrocarbon class in crude oil and reformer feed

MOLECULAR FORMULA

C_9H_{18}

9 C · one ring → C_nH_{2n}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: a ring is present, so the ring (6 C) is the parent → cyclohexane. Ethyl (2 C) and methyl (1 C) are substituents.

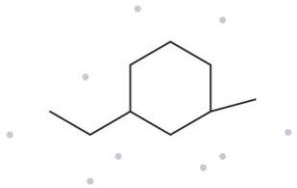
Numbering: the substituted carbons are separated by one ring carbon (the bottom one). The short way round gives 1,3 from either substituent; the long way gives 1,5. The two 1,3 options tie, so alphabetical order decides: ethyl (e) is cited first and takes C1.

Substituents: ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

3 MY NAME

1-ethyl-3-methylcyclohexane

4 THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

Returned name:

1-ethyl-3-methycyclohexane

Verdict: ✓ match

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A09

CHALLENGE

STRUCTURE → NAME

A branched substituent; the longest-chain rule still decides the parent

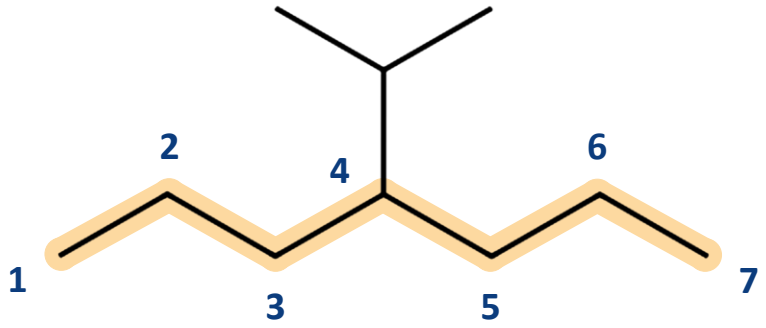
Branched C10 paraffin in the kerosene/jet-fuel range

MOLECULAR FORMULA

$C_{10}H_{22}$

10 C · open chain → C_nH_{2n+2}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: longest chain = 7 C → heptane (the horizontal zigzag). A path through the branch reaches only 6 C (CH_3–CH–C4–C5–C6–C7), so the branch is a substituent, not the parent.

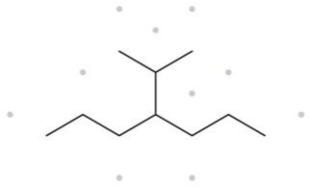
Numbering: both ends put the branch on C4 (4 either way), so there is no choice to make.

Substituents: isopropyl on C4 → 4-isopropyl. Accepted equivalents: (1-methylethyl) and propan-2-yl (Appendix B.3).

3 MY NAME

4-isopropylheptane

4 THE ALKANE CHECK



2-methyl-3-propylhexane

Returned name:

2-methyl-3-propylhexane

Verdict: ✗ mismatch

If ✗ or a different form: correct name + rule

[2-methyl-3-propylhexane → ✗: longest chain = 7 C → heptane]

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A10

CHALLENGE

STRUCTURE → NAME

A ring and a chain in one molecule: which is the parent?

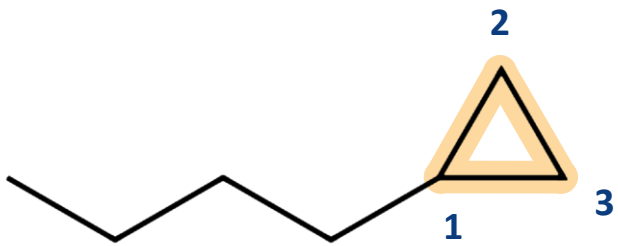
Cyclopropyl rings are common building blocks in drugs (e.g., ciprofloxacin)

MOLECULAR FORMULA

C_7H_{14}

7 C · one ring → C_nH_{2n}

1 NUMBERED PARENT



2 NAMING TRAIL

Parent: a ring is present, so the ring (3 C) is the parent → cyclopropane, even though the attached chain (4 C) is longer. This activity follows the 2013 rule: ring = parent.

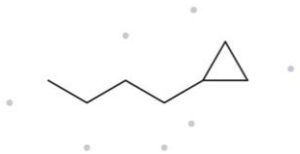
Numbering: C1 is the ring carbon that carries the chain. Both directions round the ring give the same single locant 1.

Substituents: an unbranched 4-carbon chain → butyl → butylcyclopropane (a lone ring substituent needs no locant; not 1-cyclopropylbutane).

3 MY NAME

butylcyclopropane

4 THE ALKANE CHECK



butylcyclopropane

Returned name:

butylcyclopropane

Verdict: ✓ match

If ✗ or a different form: correct name + rule

[1-cyclopropylbutane → ✗: the ring is the parent in this activity]

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

B01

FOUNDATION

NAME → STRUCTURE

GIVEN NAME

2-methylpropane

Shortest parent with one branch

Isobutane: LPG component and refrigerant R-600a

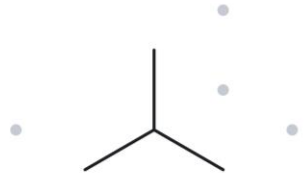
1 NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}.

2 BOND-LINE STRUCTURE (THE ALKANE)



2-methylpropane

3 FORMULA CHECK

Counted from the drawing: C_4H_{10}

Carbon types: 3 CH_3 + 1 CH → 3(3) + 1 = 10 H

Open chain: C_nH_{2n+2}, n = 4 → C_4H_{10} ✓

4 BACK-CHECK

Returned name:

2-methylpropane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B02

FOUNDATION

NAME → STRUCTURE

GIVEN NAME

3-ethylpentane

Ethyl branch on the middle carbon

C7 isomer of heptane; gasoline range

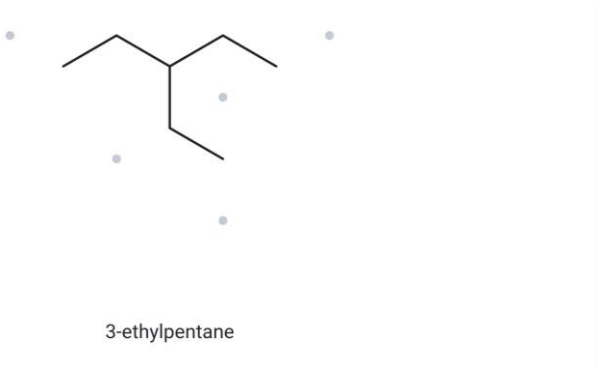
1 NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: ethyl on C3 (the middle carbon).

Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}.

2 BOND-LINE STRUCTURE (THE ALKANE)



3-ethylpentane

3 FORMULA CHECK

Counted from the drawing: C_7H_{16}

Carbon types: 3 CH_3 + 3 CH_2 + 1 CH → 9 + 6 + 1 = 16 H

Open chain: C_nH_{2n+2}, n = 7 → C_7H_{16} ✓

4 BACK-CHECK

Returned name:

3-ethylpentane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B03

FOUNDATION

NAME → STRUCTURE

GIVEN NAME

2,2-dimethylbutane

Two substituents on the same carbon

Neohexane: high-octane gasoline blending component

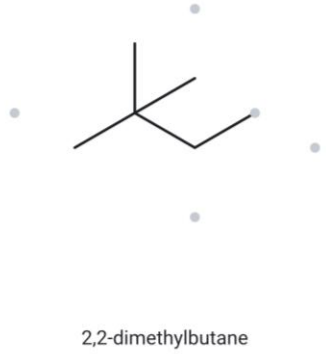
1 NAME DECODED

Parent: butane → a 4-carbon chain.

Substituents: two methyls, both on C2 (2,2-dimethyl).

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2 BOND-LINE STRUCTURE (THE ALKANE)



2,2-dimethylbutane

3 FORMULA CHECK

Counted from the drawing: C_6H_{14}

Carbon types: 4 CH_3 + 1 CH_2 + 1 C → 12 + 2 + 0 = 14 H

Open chain: C_nH_{2n+2}, n = 6 → C_6H_{14} ✓

4 BACK-CHECK

Returned name:

2,2-dimethylbutane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B04

DEVELOPING

NAME → STRUCTURE

GIVEN NAME

3-ethyl-2-methylpentane

Two different substituents on neighbouring carbons

C8 isomer of octane

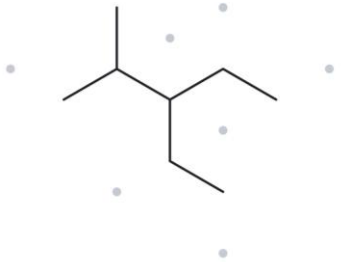
1 NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: methyl on C2; ethyl on C3 (cited alphabetically: ethyl, then methyl).

Expected formula: 5 + 1 + 2 = 8 C; open chain → C_8H_{18}.

2 BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-2-methylpentane

3 FORMULA CHECK

Counted from the drawing: C_8H_{18}

Carbon types: 4 CH_3 + 2 CH_2 + 2 CH → 12 + 4 + 2 = 18 H

Open chain: C_nH_{2n+2}, n = 8 → C_8H_{18} ✓

4 BACK-CHECK

Returned name:

3-ethyl-2-methylpentane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B05

DEVELOPING

NAME → STRUCTURE

GIVEN NAME

1-chloro-2-methylpropane

Halogen on a terminal carbon

Isobutyl chloride: alkylating agent and intermediate

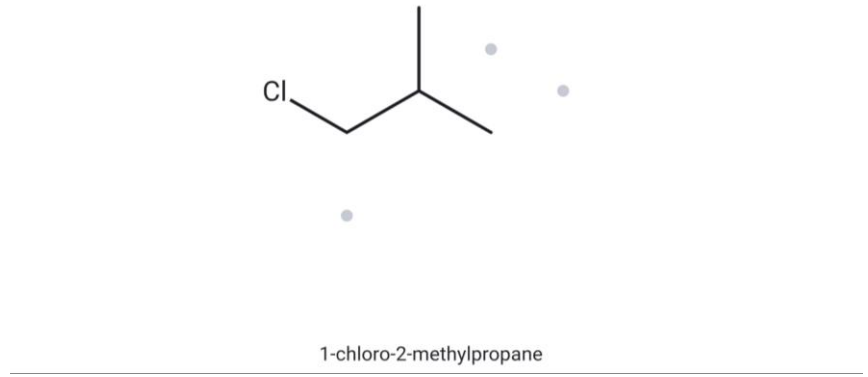
1 NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: chloro on C1; methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain C_4H_{10}; Cl replaces one H → C_4H_9Cl.

2 BOND-LINE STRUCTURE (THE ALKANE)



1-chloro-2-methylpropane

3 FORMULA CHECK

Counted from the drawing: C_4H_9Cl

Carbon types: 2 CH_3 + 1 CH_2 + 1 CH → 6 + 2 + 1 = 9 H, plus 1 Cl

Open chain: C_nH_{2n+2}, n = 4 → C_4H_{10}; Cl replaces one H → C_4H_9Cl ✓

4 BACK-CHECK

Returned name:

1-chloro-2-methylpropane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B06

DEVELOPING

NAME → STRUCTURE

GIVEN NAME

1,2-dimethylcyclopentane

Adjacent substituents on a five-membered ring

Naphthene found in petroleum naphtha

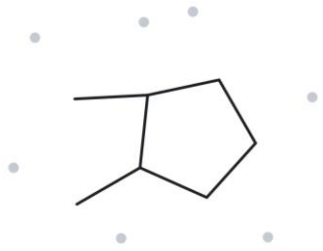
1 NAME DECODED

Parent: cyclopentane → a 5-carbon ring.

Substituents: methyl on C1 and methyl on C2 (adjacent ring carbons).

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}.

2 BOND-LINE STRUCTURE (THE ALKANE)



1,2-dimethylcyclopentane

3 FORMULA CHECK

Counted from the drawing: C_7H_{14}

Carbon types: 2 CH_3 + 3 CH_2 + 2 CH → 6 + 6 + 2 = 14 H

One ring: C_nH_{2n}, n = 7 → C_7H_{14} ✓

4 BACK-CHECK

Returned name:

1,2-dimethylcyclopentane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B07

DEVELOPING

NAME → STRUCTURE

GIVEN NAME

4-ethyl-3,3-dimethylheptane

Three substituents on a seven-carbon parent

Branched C11 paraffin in the kerosene range

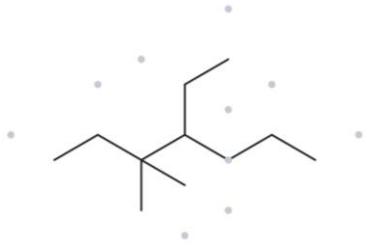
1 NAME DECODED

Parent: heptane → a 7-carbon chain.

Substituents: two methyls on C3 (3,3-dimethyl); ethyl on C4.

Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$.

2 BOND-LINE STRUCTURE (THE ALKANE)



4-ethyl-3,3-dimethylheptane

3 FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$

Carbon types: 5 CH_3 + 4 CH_2 + 1 CH + 1 C → 15 + 8 + 1 + 0 = 24 H

Open chain: C_nH_{2n+2}, n = 11 → $C_{11}H_{24}$ ✓

4 BACK-CHECK

Returned name:

4-ethyl-3,3-dimethylheptane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B08

CHALLENGE

NAME → STRUCTURE

GIVEN NAME

tert-butylcyclohexane

A common substituent name: draw it, then compare with the tool's systematic name

The bulky tert-butyl group anchors cyclohexane conformations (a later topic)

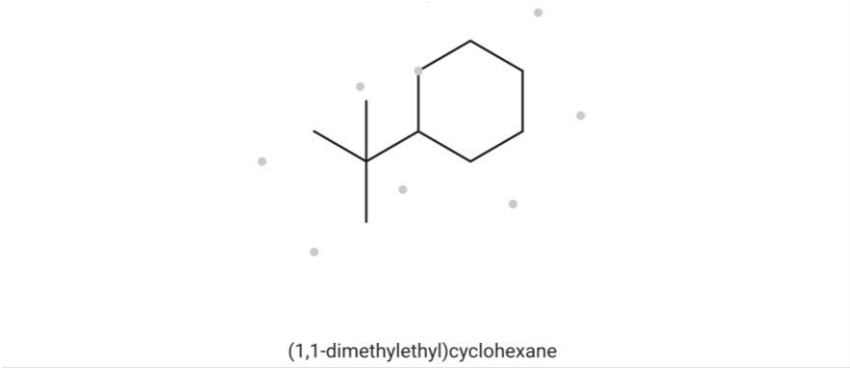
1 NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: tert-butyl, $-C(CH_3)_3$ (4 C), on C1; a lone ring substituent needs no locant. Equivalent forms: (1,1-dimethylethyl), 2-methylpropan-2-yl.

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

2 BOND-LINE STRUCTURE (THE ALKANE)



(1,1-dimethylethyl)cyclohexane

3 FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

Carbon types: 3 CH_3 + 5 CH_2 + 1 CH + 1 C → 9 + 10 + 1 + 0 = 20 H

One ring: C_nH_{2n}, n = 10 → $C_{10}H_{20}$ ✓

4 BACK-CHECK

Returned name:

(1,1-dimethylethyl)cyclohexane

If it differs from the given name, explain why:

[(1,1-dimethylethyl)cyclohexane: same compound, since tert-butyl = (1,1-dimethylethyl)]

Verdict:

✗ **differs**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B09

CHALLENGE

NAME → STRUCTURE

GIVEN NAME

3-ethyl-1,1-dimethylcyclohexane

Ring numbering with a doubly substituted carbon

Alkylcyclohexane of the kind found in jet and diesel fuel

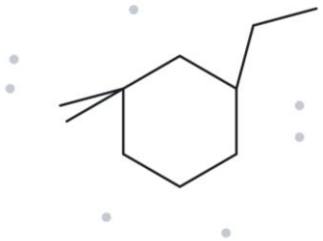
1 NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: two methyls on C1 (1,1-dimethyl); ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$.

2 BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-1,1-dimethylcyclohexane

3 FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

Carbon types: 3 CH_3 + 5 CH_2 + 1 CH + 1 C → 9 + 10 + 1 + 0 = 20 H

One ring: C_nH_{2n}, n = 10 → $C_{10}H_{20}$ ✓

4 BACK-CHECK

Returned name:

3-ethyl-1,1-dimethylcyclohexane

Verdict:

✓ **same as the given name**

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B10

CHALLENGE

NAME → STRUCTURE

GIVEN NAME

2-ethylbutane

DELIBERATELY INCORRECT NAME

This name breaks a rule: draw exactly what it says, then find the correct name

Naming errors on labels and safety data sheets cause real mix-ups

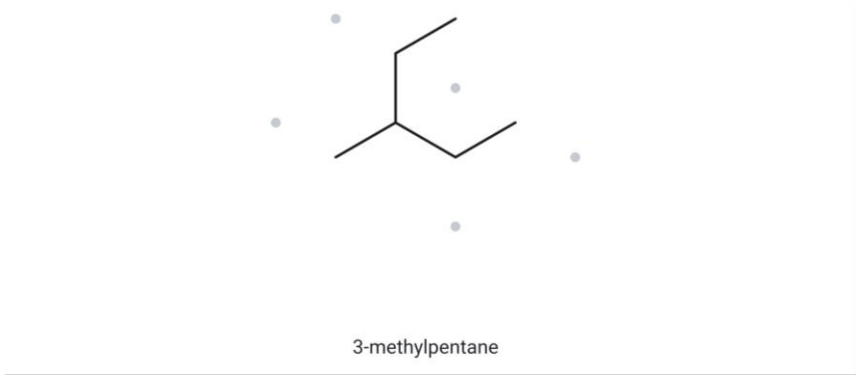
1 NAME DECODED

Parent: butane → a 4-carbon chain (as written).

Substituents: ethyl on C2 (as written).

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2 BOND-LINE STRUCTURE (THE ALKANE)



3-methylpentane

3 FORMULA CHECK

Counted from the drawing: C_6H_{14}

Carbon types: 3 CH_3 + 2 CH_2 + 1 CH → 9 + 4 + 1 = 14 H

Open chain: C_nH_{2n+2}, n = 6 → C_6H_{14} ✓

4 BACK-CHECK

Returned name:

3-methylpentane

Verdict:

✗ differs

Correct name: 3-methylpentane

Rule broken: the parent must be the longest continuous chain. As written, C4–C3–C2–CH2–CH3 runs through the ethyl group: 5 C → pentane, not butane. The old C1 becomes a methyl on C3.

Same molecule, C_6H_{14}.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded